



Penalized Logistic Regressions with Technical Indicators, Fundamental Indicators and Sentiment Indicators Predict Weekly Stock Up or Down Trends

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Abstract

Accurately predicting weekly stock up or down trends is critical in the financial markets. In order to improve the prediction accuracy, in this paper we combine technical indicators, fundamental indicators and sentiment indicators with LASSO/SCAD/MCP penalized logistic regressions for predicting weekly up or down trends of stock prices for listed companies. Firstly, 24 technical indicators, 20 fundamental indicators and 4 sentiment indicators are selected as predictor vectors, and weekly up or down trends of stock prices is used as a two-class response variable, and then construct LASSO/SCAD/MCP penalized logistic regressions. Secondly, coordinate descent algorithm and the training samples are introduced to select some important variables and get these parameter estimators, and apply the test samples and the trained models to predict weekly up or down trends of stock price. Thirdly, we calculate the confusion matrix, sensitivity and specificity, and plot the ROC curve to compare their prediction performance. Finally, we apply the proposed method to predict weekly up or down trends for four Chinese listed companies from 2010 to 2023. The prediction results show that LASSO/SCAD/MCP penalized logistic regressions with technical indicators, fundamental indicators and sentiment indicators outperform LASSO/SCAD/MCP penalized logistic regressions with technical indicators, 3 machine learning methods and 3 deep learning methods.

Keywords Weekly stock up or down trends · Technical indicators · Fundamental indicators · Sentiment indicators · Penalized logistic regressions · Coordinate descent algorithm

1 Introduction

In today's globalized financial market, the stock market is an important place for capital flows and investment decisions. Its volatility has a profound impact on investors, enterprises and the whole economic system. Stock price prediction has always been a hot and difficult problem in the economic and financial field. Existing studies mainly focus on daily or even intraday price movement prediction. For example, Kim (2003) applied support vector machine (SVM) to predict the direction of daily stock price changes of Korea Composite Stock Price Index (KOSPI); Aldin et al. (2012) applied artificial neural network (ANN) to predict the daily trend of stock price index; Patel et al. (2015) compared the performance of various machine learning models in predicting stock price movements based on daily data of 10 years; Kumar (2021) combined technical indicators with soft computing models to predict intraday price trends of individual stocks. Yang et al. (2022) combined technical analysis with group penalized logistic regression to predict daily upward and downward trends in stock prices, whereas Hu and Yang (2024a) employed group penalized trinomial logit dynamic model incorporating technical indicators to forecast daily up/sideways/down trends for stock returns. However, the short time horizon of daily forecasts cannot capture the medium and long term trend of the stock market. Different from most studies on daily trend prediction of stock prices, in this paper we mainly study the problem of weekly trend prediction of stock prices.

Technical analysis is the most commonly used method to predict the future trend of stock prices. It mainly starts from the historical behavior of price, volume, breadth and trading activity, evaluates the state of a stock using information such as charts and technical indicators, and finally predicts price trends and trading signals (Murphy, 1999; Achelis, 2001; Edwards et al., 2007; Cavalcante et al., 2016). The core of technical analysis lies in the use of charts and technical indicators to monitor the trend of price and volume, and the use of historical data on price and volume movements to predict price movements and trading signals. Oriani and Coelho (2016) evaluated the impact of different technical indicators on the prediction of stock closing prices. Vargas et al. (2018) adopted deep learning, technical indicators and financial news to forecast the stock market. Ma and Yan (2022) selected 27 technical indicators, 5 original price series, and Convolutional Neural Network (CNN) to predict stock price changes in the Chinese stock market, and so on. Hu et al. (2024) proposed semi-varying coefficient panel data model with 5 technical Indicators to predict stock returns in financial market, whereas Hu and Yang (2024b) proposed G-LASSO/G-SCAD/G-MCP penalized MLM model to exert class discovery and class prediction for high-dimensional multi-category classification problems, and combined group penalized MLM with 58 technical indicators to predict up trends, sideways trends and down trends for stock returns. However, technical analysis is mainly used for short-term forecasting and is not suitable for medium-and long-term forecasting. For example, Vanstone and Finnie (2009) found that fundamental indicators should be taken into account in the medium and long term of stock market prediction, whereas technical indicators should be taken into account in the short term forecasting (daily forecast). Therefore, it is not enough to only consider technical indicators for weekly stock

price rise and fall trend prediction, and it is more reasonable to consider both technical indicators and fundamental indicators for weekly rise and fall trend prediction.

Fundamental analysis, as the cornerstone of traditional financial analysis, provides important information about a company's financial position and economic environment (Fama & French, 1992). Collins et al. (1997) studied the correlation between book value and stock price. Lev and Zarowin (1999) studied the financial information of listed companies and found that financial information such as annual earnings, book value and cash flow had an impact on stock price fluctuations. Herawati and Putra (2018) found that stock price fluctuations are affected by factors such as debt-to-equity ratio, return on assets, current ratio, price-earnings ratio and total asset turnover. Ma et al. (2019) studied 25 financial determinants of 60 listed companies and found that financial reports could reveal part of stock value information. Phan and Chang (2024) further confirmed the potential of fundamental analysis in long-term prediction of stock market trends. Therefore, in this paper we combine technical indicators, fundamental indicators with LASSO/SCAD/MCP penalized logistic regressions to predict the weekly rise and fall trend of stock prices. In addition, recent studies have found that market sentiment and sentiment indicators, among others, also have an impact on stock price volatility.

With the rapid development of behavioral finance, more and more studies have found that market sentiment has a significant impact on stock prices. Baker and Stein (2004) used turnover rate as an indicator to measure investor sentiment in their market liquidity model, deeply discussed the correlation between trading volume and investor sentiment, and pointed out that trading volume could partly reflect investor sentiment. Baker and Wurgler (2006) believed that trading volume is a reliable yardstick to measure investors' psychological activities. Zhu and Niu (2016) pointed out that investor sentiment can affect stock prices by changing the expected growth rate of earnings and the required rate of return, and emphasized that both accounting information and investor sentiment were important factors in explaining stock prices. Audrino et al. (2020) found that sentiment variables could significantly improve volatility prediction. PH and Rishad (2020) argued that irrational emotions significantly led to excessive market fluctuations, etc.. In addition, the importance of sentiment analysis in stock market trend prediction is widely recognized. For example, Fu et al. (2020) developed a stock market crisis early warning system based on daily stock market valuation and investor sentiment indicators. Darapaneni et al. (2022) applied historical price and sentiment data to predict the future trend of stocks. Varghese and Mohan (2023) proposed a four stage model to detect the direction of information flow between the news sentiment and stock price, and the associational causal analysis identified the cause effect between the news and stock price. Cristescu et al. (2023) emphasized the key role of sentiment analysis in predicting stock market trends. Qian and Tan (2024) further confirmed the significant impact of investor sentiment on price information. Agrawal et al. (2024) combined sentiment indicators with technical analysis to predict stock market trends, providing a more comprehensive method for understanding the impact of market sentiment on stock prices. Fu and Zhang (2024) applied the two-layer LSTM model integrating sentiment indicators and technical indicators to predict stock prices, and found that the BERT LLA model had high accuracy and practical value in market sentiment analysis and stock price prediction.

Shahbandari et al. (2024) proposed a sentiment analysis method combining CNN with LSTM networks to predict stock market prices more accurately, etc.. Therefore, in this paper we comprehensively consider technical indicators, fundamental indicators and sentiment indicators, and combines penalized logistic regressions to predict the weekly rise and fall trend of stock prices.

Penalized logistic regression is a common method to predict the trend of stock price rise and fall, which has the advantages of simple calculation, high prediction accuracy, and interpretable model. In this paper, we first introduce three types of indicators: sentiment indicators, fundamental indicators and technical indicators as prediction vectors, then introduce LASSO (Tibshirani, 1996)/ SCAD (Fan & Li, 2001) / MCP (Zhang, 2010) penalty to logistic regression, and apply the training samples and coordinate descent algorithms to complete variable selection and parameter estimation, and finally use the test samples and the trained models to predict the weekly up and down trends of stock prices. Firstly, 24 technical indicators, 20 fundamental indicators and 4 sentiment indicators are selected as the prediction vector, and the weekly rise and fall trend of stock price is selected as the two-class response variable, and establish LASSO/SCAD/MCP penalized logistic regression. Secondly, the training set was used to learn model parameters and probability estimators, and the testing set and the trained models were used to calculate confusion matrix, sensitivity, specificity, precision and accuracy, and a ROC curve and the area under the ROC curve(AUC) were drawn to evaluate the prediction performance. Finally, the prediction results showed that penalized logistic regressions with technical indicators, fundamental indicators and sentiment indicators outperformed penalized logistic regressions with only technical indicators.

The main contributions of this paper are as follows:

- (1) We creatively propose LASSO/SCAD/MCP penalized logistic regression with 24 technical indicators, 20 fundamental indicators and 4 sentiment indicators to predict weekly stock up or down trends. It is insufficient for weekly stock price trends prediction to only consider technical indicators. Thus we comprehensively consider technical indicators containing short-term influence, fundamental indicators implying long-term information, and sentiments indicators including investor sentiment influence, and efficiently capture comprehensive market information.
- (2) LASSO/SCAD/MCP penalized logistic regression with technical indicators, fundamental indicators and sentiment indicators is more suitable for medium or long-term prediction. Existing researches primarily focused on the three aspects: (i) make daily or even intra-day prediction by only considering technical indicators; (ii) make long-term prediction by only considering fundamental indicators; (iii) make stock market prediction by only considering sentiment indicators. The proposed method simultaneously considers three type of indicators influencing stock market movements, more precisely shows the different roles of different indicators, and finally exhibits the better weekly prediction performance than the method only with technical indicators.

The rest of the paper is arranged as follows: Section 2 describes data sources and variable explanations. Section 3 proposes LASSO/SCAD/MCP penalized logistic regressions with technical indicators, fundamental indicators and sentiment indicators to predict the weekly up or down trend of stock price. Section 4 provides prediction performance evaluation indicators. Section 5 compares the predictive performance of different models using four data sets. Section 6 is conclusion and discussion.

Figure 1 exhibits the architecture of the new prediction method.

2 Data

2.1 Variable Explanation

The predictor indicators are composed of 24 technical indicators, 4 sentiment indicators and 20 fundamental indicators extracted from financial reports generated by the R package TTR. More details see Table 1. All the review periods followed the built-in default settings of the TTR package, which were consistent with the most common values found in technical analysis literature (e.g., Murphy, 1999; Achelis, 2001), more details see Table 2. In order to eliminate the impact of different measures, we apply the standardized forms to the predictive indicators, and introduce the coordinate descent algorithm and the R package ncvgreg to complete variable selection and obtain parameter estimators in LASSO/SCAD/MCP penalized logistic regressions.

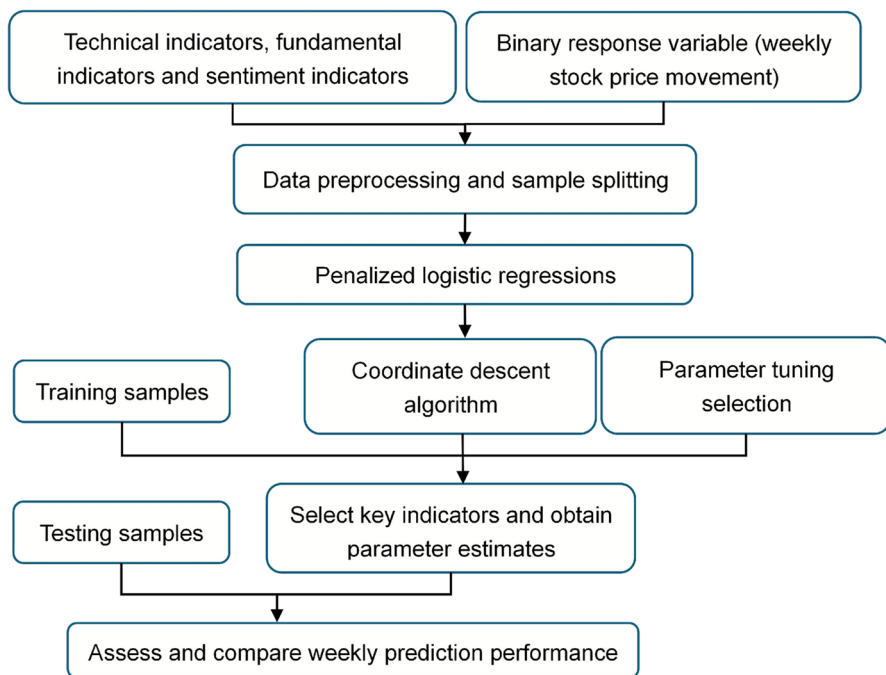


Fig. 1 The architecture of the new prediction method

Table 1 24 Technical indicators, 4 sentiment indicators and 20 fundamental indicators

	Formulae
24 Technical Indicators	
$X_{t,1}$ (WMA, Weighted Moving Average)	$WMA_t(n) = \frac{n \times C_t + (n-1) \times C_{t-1} + \dots + 1 \times C_{t-n+1}}{n \times (n+1)/2}$
$X_{t,2}$ (DEMA, Double Exponential Moving Average)	$DEMA_t(n) = 2EMA_t(n) - EMA_t(n)$
$X_{t,3}$ (SMA, Simple Moving Average)	$SMA_t(n) = \frac{C_t + C_{t-1} + \dots + C_{t-n+1}}{n}$
$X_{t,4}$ (EMA, Exponential Moving Average)	$EMA_t(n) = [2C_t + (n-1)EMA_{t-1}(n)] / (2 + EMA_{t-1}(n))$
$X_{t,5}$ (MACD, Moving Average Convergence Divergence)	$MACD_t(n) = EMA_t(12) - EMA_t(26)$
$X_{t,6}$ (RSI, Relative Strength Index)	$RSI_t(n) = 100 - 100 / \left[1 + \frac{\sum_{i=1}^n U_{t-i+1} - \sum_{i=1}^n D_{t-i+1}}{\sum_{i=1}^n U_{t-i+1} + \sum_{i=1}^n D_{t-i+1}} \right]$
$X_{t,7}$ (CMO, Chande Momentum Oscillator)	$CMO_t(n) = 100 \times \frac{\sum_{i=1}^n U_{t-i+1} - \sum_{i=1}^n D_{t-i+1}}{\sum_{i=1}^n U_{t-i+1} + \sum_{i=1}^n D_{t-i+1}}$
$X_{t,8}$ (TRIX, Triple Exponentially Smoothed Average)	$TRIX_t(n) = EMA(EMA(EMA(C_t, n_1), n_2), n_3)$
$X_{t,9}$ (DPO, Detrended Price Oscillator)	$TRIX_t(n) = 100 \times (T^3 R_t(n) / T R_{t-1}(n))$
$X_{t,10}$ (ROC, Rate of Change)	$DPO_t(n) = C_t - SMA_t(n/2 + 1)$
$X_{t,11}$ (CCI, Commodity Channel Index)	$ROC_t(n) = (C_t - C_{t-n}) / C_{t-n} \times 100$
$X_{t,12}$ (ADX, Average Directional Index)	$CCI_t(n) = (M_t - SMA_t(n)) / 0.015 D_t$
$X_{t,13}$ (Aroon, Aroon Indicator)	$SM_t = \sum_{i=1}^n M_{t-i+1} / n$, $D_t = \sum_{i=1}^n D_{t-i+1}$
	$DX_t(n) = \frac{(+D_t(n)) - (-D_t(n))}{(+D_t(n)) + (-D_t(n))}$
	$ADX_t(n) = [(n-1)ADX_{t-1}(n) + D_t] / n$
	$AroonUP_t(n) = 100 \times \frac{n - NH_t(n)}{n}$
	$AroonDW_t(n) = 100 \times \frac{n - NL_t(n)}{n}$
	$AroonI_t(n) = AroonUP_t(n) - AroonDW_t(n)$
$X_{t,14}$ (VHF, Vertical Horizontal Filter)	$VHF_t(n) = \frac{HH_t(n) - LL_t(n)}{HH_t(n) + LL_t(n)}$
$X_{t,15}$ (DC, Donchian Channel)	$DC_t(n) = [HH_t(n) + LL_t(n)] / 2$
$X_{t,16}$ (MFI, Money Flow Index)	$MFI_t(n) = 100 - 100 / \left(1 + \frac{\sum_{i=1}^n P_{t-i+1} - \sum_{i=1}^n N_{t-i+1}}{\sum_{i=1}^n P_{t-i+1} + \sum_{i=1}^n N_{t-i+1}} \right)$

Table 1 (continued)

$X_{t,17}$ (CMF,Chaikin Money Flow)	$CMF_t(n) = \sum_{i=1}^n \frac{(C_t - L_t) - (H_t - C_t)}{H_t - L_t} \cdot V_t$
$X_{t,18}$ (CAD, Chaikin Accumulation/Distribution Line)	$CAD_t(n) = CAD_{t-1} + \sum_{i=1}^n \frac{(V_{t-i} - V_{t-i-1}) \cdot (H_t - L_t)}{H_t - L_t}$
$X_{t,19}$ (OBV,On Balance Volume)	$OBV_t = OBV_{t-1} + \theta \cdot V_t, \theta = \begin{cases} +1, & \text{if } H_t > L_t \\ -1, & \text{if } H_t < L_t \end{cases}$
$X_{t,20}$ (ATR,Average True Range)	$TH_t = \max(H_t, C_{t-1}), TL_t = \min(L_t, C_{t-1})$
$X_{t,21}$ (CV,Chaikin Volatility)	$ATR_t(n) = \frac{(TH_{t-1} - TL_{t-1}) \times (n-1) + (TH_t - TL_t)}{n}$
$X_{t,22}$ (PA,Parkinson Historical Volatility)	$CV_t(n) = \frac{EMAH_{L_t} - EMAH_{L_t-n}}{EMAH_{L_t-1} \times 100}$
$X_{t,23}$ (BBands,Bollinger Bands)	$\delta_t(n) = \sqrt{\frac{N}{4n \times \log 2} \sum_{i=1}^n \frac{(H_i - L_i)^2}{L_i}}$
$X_{t,24}$ (EMV,Ease of Movement Value)	$BBands = SMA_t(\frac{H_t + L_t + C_t}{3})$
	$EMV_t = MP M_t / BR_t, BR_t = V_t / 100$
	$MP M_t = (H_t - L_t) / 2 - ((L_{t-1} - L_t -$
4 Sentiment Indicators	Formulae
$X_{t,25}$ (Change)	$Change = \frac{\text{Current Price} - \text{Previous Price}}{\text{Previous Price}}$
$X_{t,26}$ (Volume)	Number of shares traded
$X_{t,27}$ (TR, Turnover Rate)	$TR = \frac{\text{Volume}}{\text{Outstanding Shares}}$
$X_{t,28}$ (Amplitude)	$Amplitude = \frac{\text{High Price} - \text{Low Price}}{\text{Previous Close Price}}$
20 Fundamental Indicators	Formulae
$X_{t,29}$ (PE,Price-to-Earnings Ratio)	$PE = \frac{\text{Market Price per Share}}{\text{Earnings per Share(EPS)}}$
$X_{t,30}$ (PB,Price-to-Book Ratio)	$PB = \frac{\text{Market Price per Share}}{\text{Book Value per Share}}$
$X_{t,31}$ (PS,Price-to-Sales Ratio)	$PS = \frac{\text{Market Capitalization}}{\text{Total Revenue}}$
$X_{t,32}$ (PCF,Price-to-Cash Flow Ratio)	$PCF = \frac{\text{Market Price per Share}}{\text{Cash Flow per Share}}$
$X_{t,33}$ (EPS,Earnings Per Share)	$EPS = \frac{\text{Net Income} - \text{Preferred Dividend}}{\text{Number of Common Shares}}$
$X_{t,34}$ (CAR,Current Assets Ratio)	$CAR = \frac{\text{Current Assets}}{\text{Total Assets}}$
$X_{t,35}$ (FAR,Fixed Asset Ratio)	$FAR = \frac{\text{Net Fixed Assets}}{\text{Total Assets}}$

Table 1 (continued)

$X_{t,36}$ (OPM, Operating Profit Margin)	$OPM = \frac{\text{Operating Profit(EBIT)}}{\text{Revenue}}$
$X_{t,37}$ (CR, Current Ratio)	$CR = \frac{\text{Current Assets}}{\text{Current Liabilities}}$
$X_{t,38}$ (CRR, Cash Ratio)	$CRR = \frac{\text{Cash and Cash Equivalents}}{\text{Current Liabilities}}$
$X_{t,39}$ (QR, Quick Ratio)	$QR = \frac{\text{Current Assets} - \text{Inventories}}{\text{Current Liabilities}}$
$X_{t,40}$ (DAR, Debt-to-Asset Ratio)	$DAR = \frac{\text{Total Liabilities}}{\text{Total Assets}}$
$X_{t,41}$ (TAG, Year-over-Year Growth Rate of Total Assets)	$TAG = \frac{\text{Total Assets}(\text{Current Year}) - \text{Total Assets}(\text{Previous Year})}{\text{Total Assets}(\text{Previous Year})}$
$X_{t,42}$ (TPG, Year-over-Year Growth Rate of Total Profit)	$TPG = \frac{\text{Total Profit}(\text{Current Year}) - \text{Total Profit}(\text{Previous Year})}{\text{Total Profit}(\text{Previous Year})}$
$X_{t,43}$ (TRG, Year-over-Year Growth Rate of Total Revenue)	$TRG = \frac{\text{Total Revenue}(\text{Current Year}) - \text{Total Revenue}(\text{Previous Year})}{\text{Total Revenue}(\text{Previous Year})}$
$X_{t,44}$ (NAG, Year-over-Year Growth Rate of Net Assets)	$NAG = \frac{\text{Net Assets}(\text{Current Year}) - \text{Net Assets}(\text{Previous Year})}{\text{Net Assets}(\text{Previous Year})}$
$X_{t,45}$ (TAT, Total Asset Turnover)	$TAT = \frac{\text{Revenue}(\text{Total Sales})}{\text{Average Total Assets}}$
$X_{t,46}$ (IT, Inventory Turnover)	$IT = \frac{\text{Cost of Goods Sold(COGS)}}{\text{Average Inventory}}$
$X_{t,47}$ (ART, Accounts Receivable Turnover)	$ART = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$
$X_{t,48}$ (ROE, Return on Equity)	$ROE = \frac{\text{Net Income}}{\text{Average Shareholders' Equity}}$

Note: C_t represents the closing price, H_t represents the highest price, L_t represents the lowest price, $M_t = (H_t + L_t + C_t) / 3$ represents the typical price, V_t represents the trading volume, $+DI_t(n) (-DI_t(n))$ is the percentage of the true range that is up(down), $NH_t(n) (NL_t(n))$ is the number of days from the highest(lowest) price in n days to today, $HH_t(n) (LL_t(n))$ represents the highest (lowest) value over n days, $PMF_t (NMF_t)$ is the positive(negative) money flow, $EMAH L_t$ is the n period exponential moving average of $H_t - L_t$, $UP_t (DW_t)$ means upward(downward) price change at time t , N represents the period number, n is the default number of cycles in the TTR package

Table 2 Technical indicators and their default look-back windows in TTR package

Technical indicators	Window (weeks)	Technical indicators	Window (weeks)	Technical indicators	Window (weeks)
$X_{t,1}(WMA)$	10	$X_{t,9}(DPO)$	10	$X_{t,17}(CMF)$	20
$X_{t,2}(DEMA)$	10	$X_{t,10}(ROC)$	1	$X_{t,18}(CAD)$	None
$X_{t,3}(SMA)$	10	$X_{t,11}(CCI)$	20	$X_{t,19}(OBV)$	None
$X_{t,4}(EMA)$	10	$X_{t,12}(ADX)$	14	$X_{t,20}(ATR)$	14
$X_{t,5}(MACD)$	12 / 26 / 9	$X_{t,13}(Aroon)$	20	$X_{t,21}(CV)$	10
$X_{t,6}(RSI)$	14	$X_{t,14}(VHF)$	28	$X_{t,22}(PA)$	10
$X_{t,7}(CMO)$	14	$X_{t,15}(DC)$	10	$X_{t,23}(BBands)$	20
$X_{t,8}(TRIX)$	20 (signal 9)	$X_{t,16}(MFI)$	14	$X_{t,24}(EMV)$	9

2.2 Data Sources

We choose the weekly stock price data from 1 January 2010 to 31 December 2023 as the total sample for four Chinese listed companies: Guangdong Shaoneng Group Co., Ltd.(Shaoneng, stock code: 000601), Jiangsu Hengrui Medicine Co.,Ltd.(Hengrui, stock code: 600276), Poly Developments and Holdings Group Co., Ltd. (Poly, stock code: 600048) and Foshan Electrical and Lighting Co.,Ltd.(Foshan, stock code: 000541). Although the weekly sampling yields fewer weekly data than daily data, the weekly samples spanned more than 14 years and included the highly volatile COVID-19 period, ensuring adequate coverage for statistical learning. The sample set is divided into two parts: the first 70% sample is used as a training set to learn the LASSO/SCAD/MCP penalized logistic regressions, and the remaining 30% sample is used as a testing set to evaluate the prediction performance. This study focuses on the weekly rise or fall of stock prices. Fundamental indicator data belongs to quarterly data. We need process the quarterly data to match the time frequency of the weekly data so that the quarterly fundamental indicators released at the end of each quarter are directly assigned to all trading weeks within that quarter. Each stock is taken from the Oriental Wealth Choice data. Their closing price trend is shown in Fig. 2. Due to the stock market suspension and the periodic nature of the technical indicators (leading missing values: TRIX-20 has a maximum of 58 weeks; lagging missing values: DPO-10 is 6 weeks), we removed a total of 64 non-overlapping observations and retained over 90% of the original observations without representative loss. The training sets and the testing sets for the four Chinese listed companies is shown in Table 3 after excluding the missing data.

3 LASSO/SCAD/MCP Penalized Logistic Regressions

Assume that a binary response variable

$$Y_t = \begin{cases} 1, & C_{t+1} > C_t, \\ 0, & C_{t+1} \leq C_t, \end{cases} \quad (1)$$

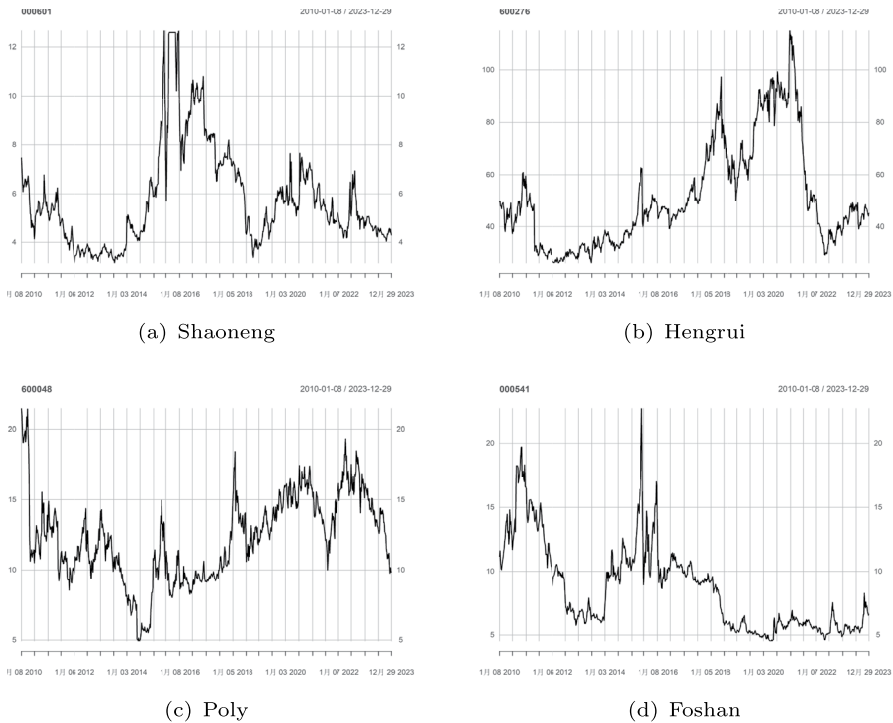


Fig. 2 Stock closing price trends from 1 January 2010 to 31 December 2023

Table 3 The training set and the testing sets for the four Chinese listed companies

Data set	The training set			The testing set		
	Number of rises	Number of declines	Total	Number of rises	Number of declines	Total
Shaoneng	220	226	446	89	102	191
Hengrui	253	203	456	96	99	195
Poly	229	227	456	86	109	195
Foshan	222	232	454	93	101	194

represents stock price movement trends, where $Y_t = 1$ indicates an upward trend, and $Y_t = 0$ indicates a downward trend. The prediction vector $X_t = (X_{t,1}, \dots, X_{t,48})^T$ consists of 48 indicators that can influence the trend of stock price movements. The training set $\{(X_t, Y_t), t = 1, \dots, n_1\}$ consists of a binary response variable $Y_t \in \{0, 1\}$ and a predictor vector $X_t \in \mathbb{R}^{48}$. Then, logistic regression is defined as

$$p_{\beta}(X_t) = P_{\beta}(Y_t = 1|X_t) = \frac{\exp\left(\beta_0 + \sum_{j=1}^{48} X_{t,j}\beta_j\right)}{1 + \exp\left(\beta_0 + \sum_{j=1}^{48} X_{t,j}\beta_j\right)}, \quad (2)$$

$$1 - p_{\beta}(X_t) = P_{\beta}(Y_t = 0|X_t) = \frac{1}{1 + \exp\left(\beta_0 + \sum_{j=1}^{48} X_{t,j} \beta_j\right)}, \quad (3)$$

where $\beta = (\beta_0, \beta_1, \dots, \beta_{48})^T$ is an unknown parameter vector, $X_t = (X_{t,1}, \dots, X_{t,48})^T$ is the prediction vector, and β_0 is the intercept term. Based on the training set $\{(X_t, Y_t), t = 1, \dots, n_1\}$, the log-likelihood function to logistic regression becomes

$$l(\beta) = \sum_{t=1}^{n_1} \{Y_t \eta_{\beta}(X_t) - \log[1 + \exp(\eta_{\beta}(X_t))]\}, \quad (4)$$

where $\eta_{\beta}(X_t) = \log\left\{\frac{p_{\beta}(X_t)}{1 - p_{\beta}(X_t)}\right\} = \beta_0 + \sum_{j=1}^{48} X_{t,j} \beta_j$, and the penalized negative log-likelihood function can be rewritten as

$$S_{\lambda, \gamma}(\beta) = -l(\beta) + \sum_{j=1}^{48} P(\beta_j; \lambda, \gamma), \quad (5)$$

where $P(\beta_j; \lambda, \gamma)$ is the penalty function listed in Table 4, λ is the tuning parameter and γ is the regularization parameter. Equation (5) does not penalize the intercept β_0 , and the estimate that minimizes $S_{\lambda, \gamma}(\beta)$ is denoted as $\hat{\beta}$. According to Hu and Liu (2020), parameter estimators in Table 5 can be obtained using coordinate descent algorithm.

4 Two-Class Prediction Performance

4.1 Two-Class Confusion Matrix

The two-class confusion matrix (see Table 6) is a cross-table between the actual class and the predicted class, where TP represents correctly predicted price increase, FP represents a price decrease incorrectly predicted as an increase, FN represents a price increase incorrectly predicted as a decrease, and TN represents correctly predicted price decrease. Accuracy, precision, sensitivity and specificity are calculated accord-

Table 4 LASSO/SCAD/MCP penalties

Signs	3 penalty functions
LASSO	$P_{LASSO}(\beta; \lambda) = \lambda \sum_{j=1}^p \beta_j , \lambda > 0$
SCAD	$P_{SCAD}(\beta; \lambda, \gamma) = \begin{cases} \lambda \gamma \beta - 0.5(\frac{\lambda^2}{\gamma} - \lambda \beta) & \text{if } \lambda \gamma \beta \leq 0.5(\frac{\lambda^2}{\gamma} - \lambda \beta) \\ \frac{\lambda^2}{2\gamma} & \text{otherwise} \end{cases}$
MCP	$P_{MCP}(\beta; \lambda, \gamma) = \begin{cases} \lambda \gamma \beta - 0.5(\frac{\lambda^2}{\gamma} - \lambda \beta) & \text{if } \lambda \gamma \beta \leq 0.5(\frac{\lambda^2}{\gamma} - \lambda \beta) \\ \frac{1}{2} \gamma \lambda^2 & \text{otherwise} \end{cases}$

Table 5 Parameter estimation for penalized logistic regression

Models	Parameter estimators
LASSO	$\hat{\beta}_j^{LASSO} = \frac{S(Z_j, \lambda)}{v_j}$
SCAD	$\hat{\beta}_j^{SCAD} = \begin{cases} \frac{S(Z_j, \lambda)}{v_j}, & Z_j \leq v_j \lambda \\ \frac{S(Z_j, \gamma \lambda / (\gamma - 1))}{v_j - 1 / (\gamma - 1)}, & \lambda(v_j - 1 / (\gamma - 1)) < Z_j \leq v_j \lambda \end{cases}$
MCP	$\hat{\beta}_j^{MCP} = \begin{cases} \frac{S(Z_j, \lambda)}{v_j + 1 / \gamma}, & Z_j \leq v_j \lambda \\ \frac{Z_j}{v_j}, & Z_j > v_j \lambda \end{cases}$
notation	$v_j = n_1^{-1} X_j^\top W X_j, j = 1, \dots, p, Z_j = n_1^{-1} X_j^\top \tilde{Y}$ $\tilde{Y} = X^\top \beta^{(m)} + W^{-1}(Y - P), X = (X_1, \dots, X_p)^\top$ $\beta_{-j} = (\beta_1, \dots, \beta_{j-1}, 0, \beta_{j+1}, \dots, \beta_p)^\top$

Note: $X = (X_1, \dots, X_{n_1})^\top$ is an $n_1 \times p$ matrix of predictor variables, $Y = (Y_1, \dots, Y_{n_1})^\top$ is an $n_1 \times 1$ vector of response variables, W is a weighted diagonal matrix, P is the current iteration estimation probability, $\beta^{(m)}$ is the current iteration parameter, n_1 is the sample size of the training set, and p is the number of predictor variables

Table 6 Two-class confusion matrix

	Observe up: $Y_t = 1$	Observe down: $Y_t = 0$
Predict up: $\hat{Y}_t = 1$	TP	FP
Predict down: $\hat{Y}_t = 0$	FN	TN

ing to Table 6 to evaluate the performance of stock price prediction for upward and downward movements.

According to Table 6, one can compute accuracy, precision, sensitivity and specificity to evaluate stock price prediction performance, where accuracy refers to the proportion of correctly predicted overall samples, i.e.,

$$accuracy = \frac{TP + TN}{TP + FP + FN + TN},$$

precision measures the proportion of real positive samples in the predicted positive samples, i.e.,

$$precision = \frac{TP}{TP + FP},$$

sensitivity measures the ability to correctly identify positive samples or up trends, i.e.,

$$sensitivity = \frac{TP}{TP + FN},$$

specificity measures the ability to correctly identify negative samples or down trends, i.e.,

$$specificity = \frac{TN}{TN + FP}.$$

4.2 ROC Curve

The ROC curve is plotted by specificity and sensitivity, where X axis is 1-specificity (FPR) and Y axis is sensitivity (TPR). The closer the ROC curve is to the upper left corner, the better the performance of the classifier. In the upper left corner, TPR is close to 1, whereas FPR is close to 0. This implies that the classifier correctly identifies most of the positive samples but misclassifying fewer negative samples. AUC is an important metric for evaluating classifier performance. The higher the AUC value, the better the performance of the classifier. Typically, the AUC value ranges from 0.5 to 1, where 0.5 indicates that the performance of the classifier is comparable to a random guess, and 1 indicates that the classifier performs perfectly.

5 Empirical Analysis

5.1 Model Estimators

Parameter estimators for LASSO/SCAD/MCP penalized logistic regressions can be computed using the training set and the coordinate descent algorithm. For LASSO/SCAD/MCP penalized logistic regressions, λ controls the strength of the penalty, the coefficients gradually converging to 0 as λ increases. When analyzing different companies, we found that for Shaoneng data, the SCAD penalty performs better at $\gamma=3.5$, whereas the MCP penalty performs better at $\gamma=5$; for Hengrui and Poly data, both SCAD and MCP penalties perform better at $\gamma=10$; whereas for Foshan data, both SCAD and MCP perform better at $\gamma=5$. Figures 3, 4, 5, and 6 show the coefficient path diagrams for different models with the four datasets.

The relatively optimal λ values selected using 10-fold cross-validation and training set selection are shown in Figures 7, 8, 9, and 10, where the vertical coordinate represents the cross-validation error, the horizontal coordinate represents $\log(\lambda)$, and the numbers above indicate the number of variables selected by the model at the corresponding λ . Table 7 lists the relative optimal λ for LASSO/SCAD/MCP penalized logistic regressions based on the four datasets.

Coordinate descent algorithm is applied to LASSO/SCAD/MCP penalized logistic regressions with the optimal λ , thus complete variable selection and obtain parameter estimator. The selected predictor variables are shown in Table 8. LASSO/SCAD/MCP penalized logistic regression with the relative optimal λ or γ is used to predict upward or downward trend:

1. Shaoneng data set

(A) LASSO penalized logistic regression

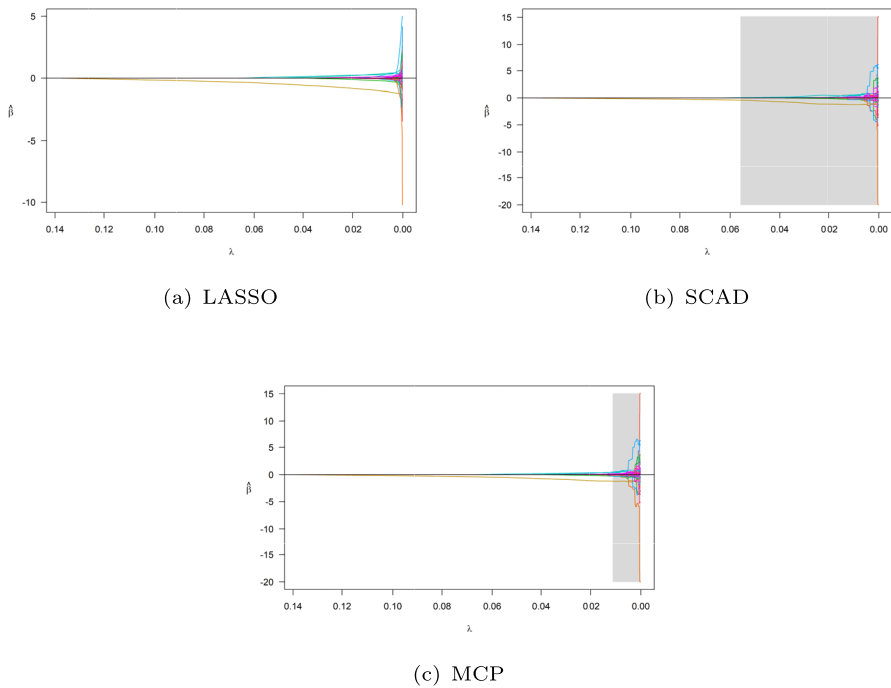


Fig. 3 Coefficient path diagram for the Shaoneng dataset

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,1}}}{1 + e^{Z_{t,1}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,1}}}, \quad (6)$$

where $Z_{t,1} = -0.0494 - 0.8675X_{t,9} - 0.1384X_{t,17} + 0.2368X_{t,24} + 0.2442X_{t,25} + 0.0302X_{t,26} + 0.0590X_{t,36} + 0.0774X_{t,42} - 0.0260X_{t,46}$

(B) SCAD penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,2}}}{1 + e^{Z_{t,2}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,2}}}, \quad (7)$$

where $Z_{t,2} = -0.0371 - 0.7163X_{t,9} + 0.0792X_{t,24} + 0.1581X_{t,25}$.

(C) MCP penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,3}}}{1 + e^{Z_{t,3}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,3}}}, \quad (8)$$

where

$Z_{t,3} = -0.0471 - 0.9043X_{t,9} - 0.0702X_{t,17} + 0.1416X_{t,24} + 0.2901X_{t,25} + 0.0302X_{t,42}$

2. Hengrui data set

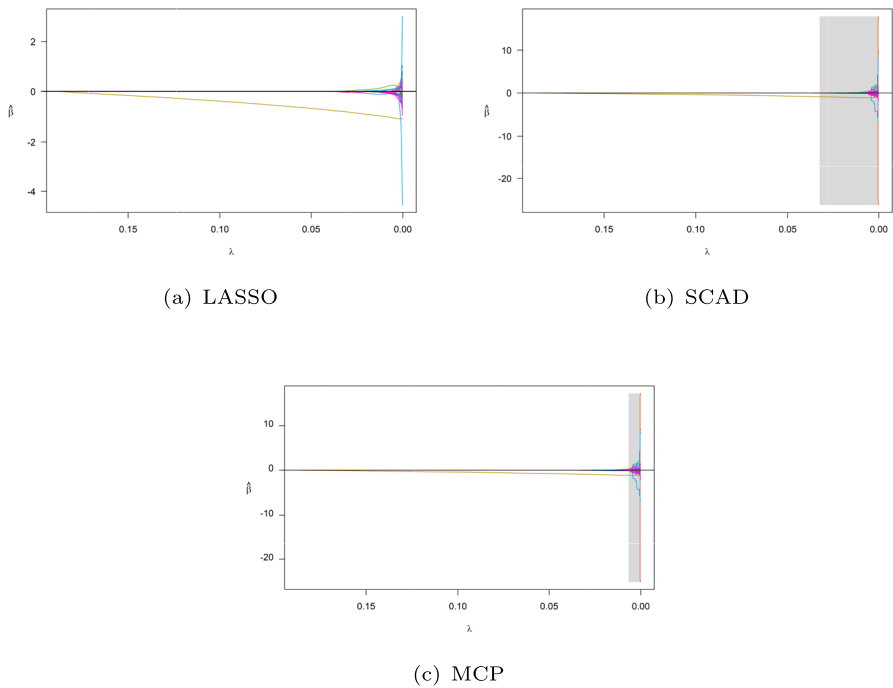


Fig. 4 Coefficient path diagram for the Hengrui dataset

(A) LASSO penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,4}}}{1 + e^{Z_{t,4}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,4}}}, \quad (9)$$

where

$$Z_{t,4} = 0.2299 - 0.8244X_{t,9} + 0.0273X_{t,12} - 0.0172X_{t,33} - 0.0217X_{t,45}.$$

(B) SCAD penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,5}}}{1 + e^{Z_{t,5}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,5}}}, \quad (10)$$

where

$$Z_{t,5} = 0.2316 - 0.9132X_{t,9} + 0.0377X_{t,12} - 0.0362X_{t,33} - 0.0126X_{t,45}.$$

(C) MCP penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,6}}}{1 + e^{Z_{t,6}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,6}}}, \quad (11)$$

where $Z_{t,6} = 0.2280 - 0.8546X_{t,9} - 0.0104X_{t,45}.$

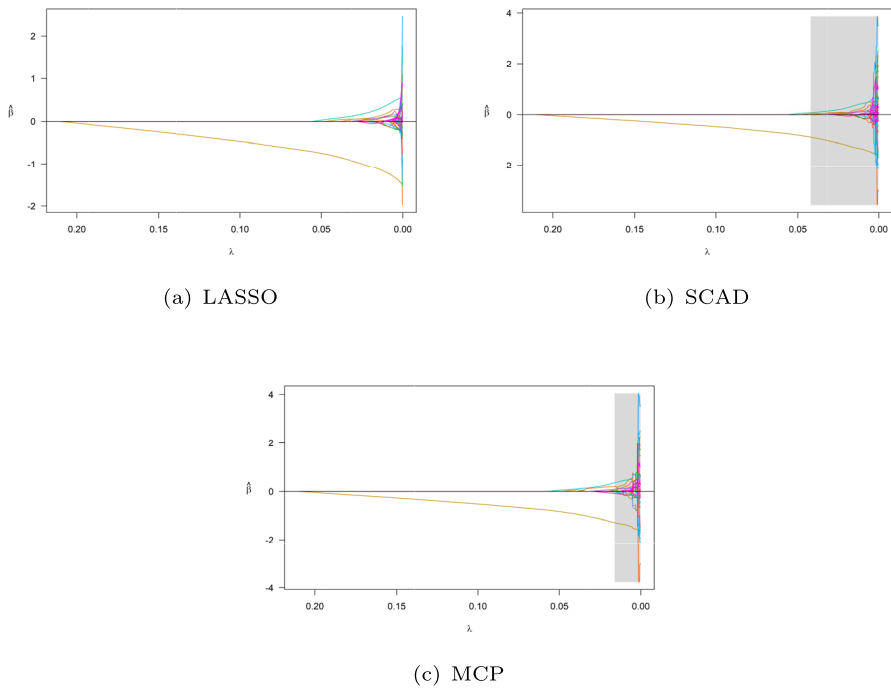


Fig. 5 Coefficient path diagram for the Poly dataset

3. Poly data set

(A) LASSO penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,7}}}{1 + e^{Z_{t,7}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,7}}}, \quad (12)$$

where $Z_{t,7} = -0.0046 + 0.0911X_{t,2} + 0.0251X_{t,6} - 1.1496X_{t,9} + 0.1350X_{t,11} + 0.0682X_{t,14} - 0.0378X_{t,16} + 0.3038X_{t,24} - 0.0353X_{t,28} + 0.0877X_{t,36} - 0.0337X_{t,45} - 0.0500X_{t,48}$

(B) SCAD penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,8}}}{1 + e^{Z_{t,8}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,8}}}, \quad (13)$$

where $Z_{t,8} = -0.0037 + 0.0503X_{t,2} + 0.0744X_{t,6} - 1.1987X_{t,9} + 0.0819X_{t,11} + 0.0195X_{t,14} + 0.2632X_{t,24} - 0.0021X_{t,28} + 0.0575X_{t,36} - 0.0533X_{t,45} - 0.0004X_{t,48}$

(C) MCP penalized logistic regression

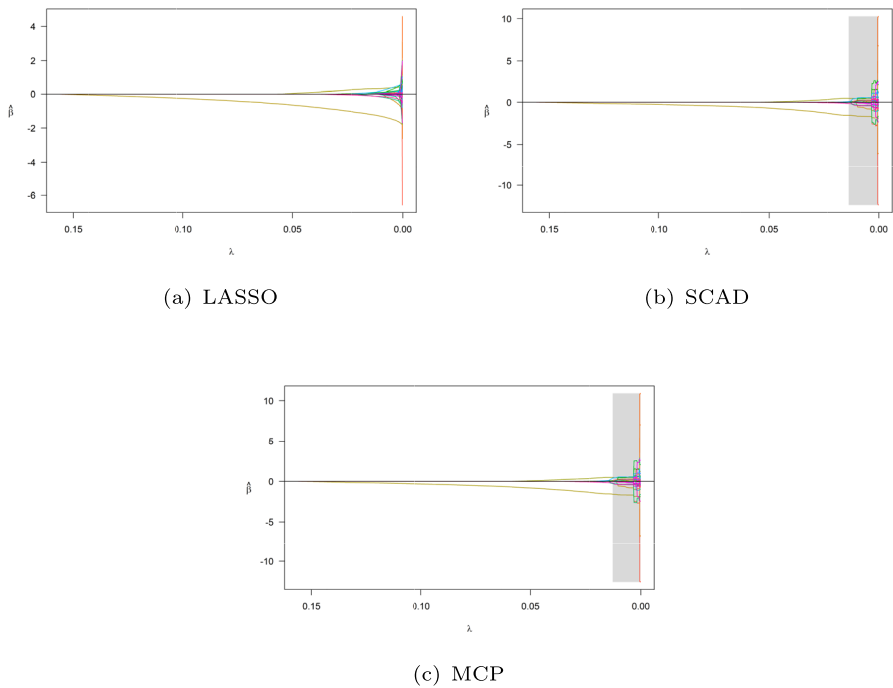


Fig. 6 Coefficient path diagram for the Foshan dataset

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,9}}}{1 + e^{Z_{t,9}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,9}}}, \quad (14)$$

where

$$Z_{t,9} = -0.0042 + 0.0656X_{t,2} - 1.1897X_{t,9} + 0.1461X_{t,11} + 0.0244X_{t,14} + 0.2647X_{t,24} + 0.0534X_{t,36} - 0.0533X_{t,45}$$

4. Foshan data set

(A) LASSO penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,10}}}{1 + e^{Z_{t,10}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,10}}}, \quad (15)$$

where

$$Z_{t,10} = -0.0847 - 1.0278X_{t,9} + 0.2359X_{t,10} + 0.0265X_{t,14} + 0.0067X_{t,28} - 0.0546X_{t,46}$$

(B) SCAD penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,11}}}{1 + e^{Z_{t,11}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,11}}}, \quad (16)$$

where

$$Z_{t,11} = -0.0932 - 1.1864X_{t,9} + 0.2782X_{t,10} + 0.0052X_{t,14} - 0.0380X_{t,46}$$

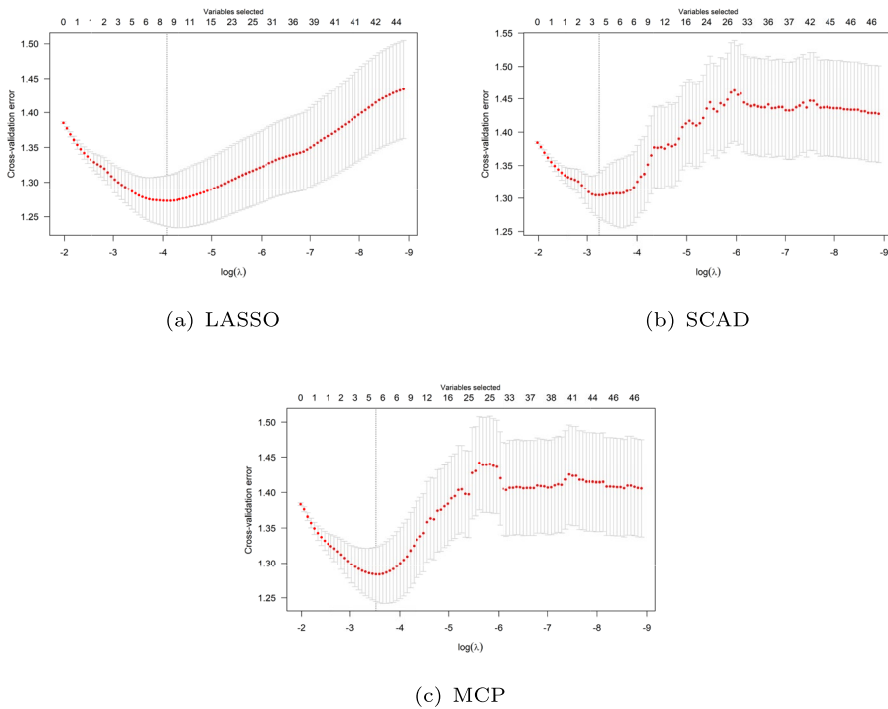


Fig. 7 Cross-validation errors in the Shaoneng dataset

(C) MCP penalized logistic regression

$$\hat{P}(Y_t = 1 | X_t) = \frac{e^{Z_{t,12}}}{1 + e^{Z_{t,12}}}, \hat{P}(Y_t = 0 | X_t) = \frac{1}{1 + e^{Z_{t,12}}}, \quad (17)$$

where

$Z_{t,12} = -0.0997 - 1.3499X_{t,9} + 0.3747X_{t,10} + 0.0152X_{t,14} - 0.0626X_{t,46}$
 From Table 8 and models (6)-(17), we observe that among technical indicators, $X_{t,9}$ (*DPO*) ranks first in its impact on stock price trends, $X_{t,14}$ (*VHF*) and $X_{t,24}$ (*EMV*) rank second, $X_{t,2}$ (*DEMA*), $X_{t,10}$ (*ROC*) and $X_{t,11}$ (*CCI*) rank third, the remaining technical indicators are much less influential. Among sentiment indicators, $X_{t,25}$ (*Change*) and $X_{t,28}$ (*Amplitude*) rank first, $X_{t,26}$ (*Volume*) ranks second, and $X_{t,27}$ (*TR*) ranks third. Among fundamental indicators, $X_{t,45}$ (*TAT*) ranks first, $X_{t,36}$ (*OPM*) and $X_{t,46}$ (*IT*) rank second, $X_{t,33}$ (*EPS*), $X_{t,42}$ (*TPG*) and $X_{t,48}$ (*ROE*) rank third, all other fundamental indicators have comparatively minor effects.

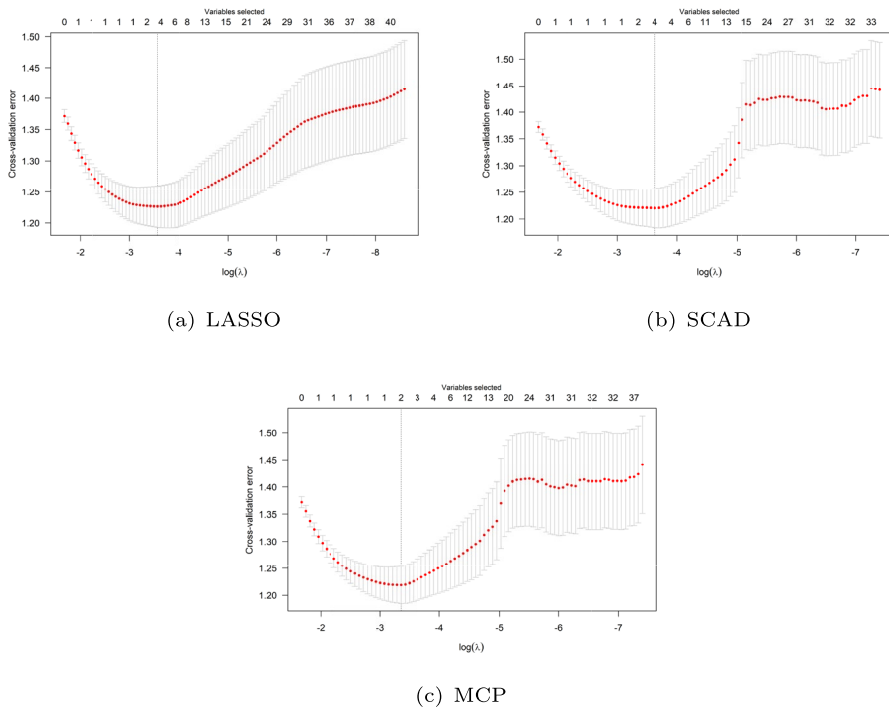


Fig. 8 Cross-validation errors in the Hengrui dataset

5.2 Two-Class Predictive Performance

Assume that $n_2 = n - n_1$ is the sample size of the testing set. Substitute the testing set $\{(x_t, Y_t), t = n_1 + 1, \dots, n\}$ into model (6)-(17), we calculate the probability estimators. For a given threshold c ,

$$\text{If } \hat{P} > c, \text{ then } \hat{Y}_t = 1; \text{ otherwise, } \hat{Y}_t = 0. \quad (18)$$

For balanced data, $c = 0.5$ is generally used; for unbalanced data, the Youden index (Raghavan et al., 2016) is often used to select c . Table 9 lists the optimal thresholds selected by the Youden index for the four datasets. Given the relative optimal threshold c , we can estimate the predicted value $\hat{Y}_t$.

In order to further demonstrate the improved predictive performance of the proposed method, we compare the predictive performance of LASSO/SCAD/MCP penalized logistic regression with three types of indicators with that of LASSO/SCAD/MCP penalized logistic regression only including technical indicators, machine learning methods (RF/SVM/ANN) or deep learning methods (DNN/CNN/LSTM) including three types of indicators. Tables 10 and 11 present the hyperparameter space settings for the three machine learning methods and the three deep learning methods, respectively. The deep learning models employ a hybrid search space: a 20% probability of using manually tuned fixed values and an 80% probability of ran-

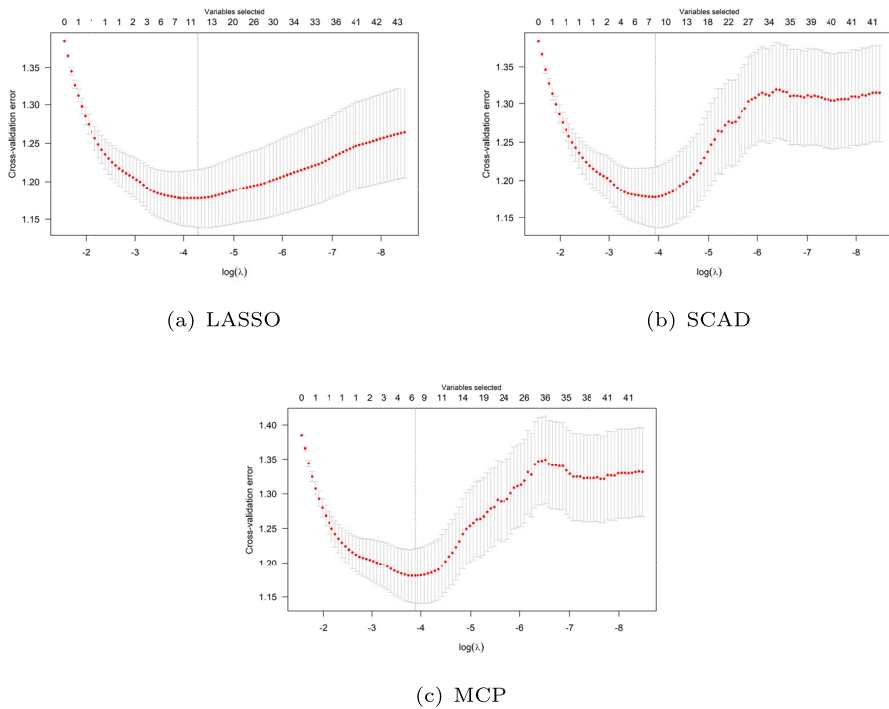


Fig. 9 Cross-validation errors in the Poly dataset

domly searching hyper-parameters within the ranges listed in these tables. Table 12 lists the results of the random search optimization. Table 13 compares the predictive performance (Sensitivity, Specificity, Precision, Accuracy) of the 12 methods based on four listed companies. From Table 13 we observe the conclusions: In terms of accuracy, penalized logistic regression models incorporating three types of indicators (LASSO/SCAD/MCP) rank first (accuracy rates generally exceeding 71%). Models using only technical indicators (LASSO_24/SCAD_24/MCP_24) rank second, the three machine learning methods (RF/SVM/ANN) rank third, whereas the three deep learning approaches (DNN/CNN/LSTM) rank fourth. Therefore, the proposed methods outperform the other 9 methods.

Figures 11, 12, 13, and 14 show the ROC curves to the 12 methods for the four datasets, where the AUC values (generally surpassing 0.75) for the 6 statistical learning methods (LASSO/SCAD/MCP/LASSO_24/SCAD_24/MCP_24) exceed those of the three machine learning methods and the three deep learning methods. However, the AUC values of LASSO/SCAD/MCP penalized logistic regression with three types of indicators are not significantly different from those of LASSO/SCAD/MCP penalized logistic regression using only technical indicators. From Table 13 and Figures 11, 12, 13, and 14, we observe that LASSO/SCAD/MCP penalized logistic regression incorporating three types of indicators outperforms LASSO/SCAD/MCP penalized logistic regression using only technical indicators, the three machine

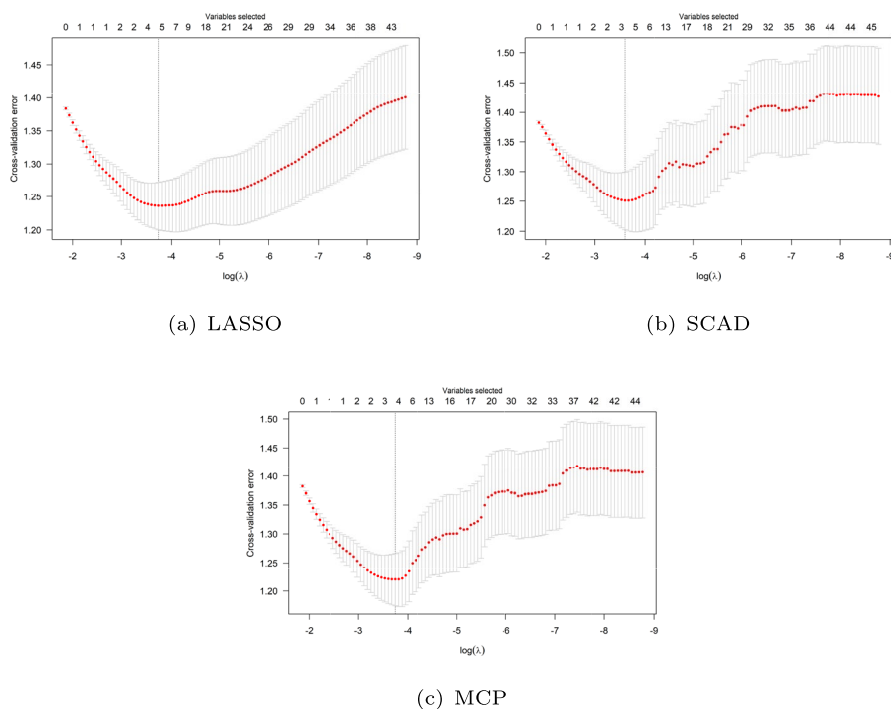


Fig. 10 Cross-validation errors in the Foshan dataset

Table 7 The relative optimal λ for the four dataset choices

	λ_{LASSO}	λ_{SCAD}	λ_{MCP}
Shaoneng	0.0170	0.0393	0.0297
Hengrui	0.0284	0.0265	0.0351
Poly	0.0138	0.0196	0.0210
Foshan	0.0237	0.0272	0.0237

learning methods(RF/SVM/ANN) and the three deep learning methods(DNN/CNN/LSTM) in terms of accuracy and AUC.

6 Conclusion and Discussion

6.1 Conclusion

In this paper we combine technical indicators, fundamental indicators, and sentiment indicators with LASSO/SCAD/MCP penalized logistic regression to predict weekly upward and downward stock price trends. We compare the predictive performance of the 12 methods, and show that LASSO/SCAD/MCP penalized logistic regressions with three types of indicators outperform the LASSO/SCAD/MCP penalized logistic regression only with technical indicators, the 3 machine learning methods and the 3

Table 8 Variable selection for LASSO/SCAD/MCP penalized logistic regression

Indicators	Shaoneng			Hengrui			Poly			Foshan		
	LASSO	SCAD	MCP	LASSO	SCAD	MCP	LASSO	SCAD	MCP	LASSO	SCAD	MCP
$X_{t,1}(WMA)$												
$X_{t,2}(DEMA)$							✓	✓	✓			
$X_{t,3}(SMA)$												
$X_{t,4}(EMA)$												
$X_{t,5}(MACD)$												
$X_{t,6}(RSI)$							✓	✓				
$X_{t,7}(CMO)$												
$X_{t,8}(TRIX)$												
$X_{t,9}(DPO)$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$X_{t,10}(ROC)$										✓	✓	✓
$X_{t,11}(CCI)$							✓	✓	✓			
$X_{t,12}(ADX)$				✓	✓							
$X_{t,13}(Aroon)$												
$X_{t,14}(VHF)$							✓	✓	✓	✓	✓	✓
$X_{t,15}(DC)$												
$X_{t,16}(MFI)$							✓					
$X_{t,17}(CMF)$	✓		✓									
$X_{t,18}(CAD)$												
$X_{t,19}(OBV)$												
$X_{t,20}(ATR)$												
$X_{t,21}(CV)$												
$X_{t,22}(PA)$												
$X_{t,23}(BBands)$												
$X_{t,24}(EMV)$	✓	✓	✓				✓	✓	✓			✓
$X_{t,25}(Change)$	✓	✓	✓									

Table 8 (continued)

Indicators	Shaoneng			Hengrui			Poly			Foshan		
	LASSO	SCAD	MCP	LASSO	SCAD	MCP	LASSO	SCAD	MCP	LASSO	SCAD	MCP
$X_{t,26}(Volume)$												
$X_{t,27}(TR)$												
$X_{t,28}(Amplitude)$							✓	✓		✓		
$X_{t,29}(PE)$												
$X_{t,30}(PB)$												
$X_{t,31}(PS)$												
$X_{t,32}(PCF)$					✓							
$X_{t,33}(EPS)$				✓								
$X_{t,34}(CAR)$												
$X_{t,35}(FAR)$												
$X_{t,36}(OPM)$ ✓							✓	✓	✓			
$X_{t,37}(CR)$												
$X_{t,38}(CRR)$												
$X_{t,39}(QR)$												
$X_{t,40}(DAR)$												
$X_{t,41}(TAG)$												
$X_{t,42}(TPG)$ ✓			✓									
$X_{t,43}(TRG)$												
$X_{t,44}(NAG)$												
$X_{t,45}(TAT)$				✓	✓	✓	✓	✓	✓	✓	✓	✓
$X_{t,46}(IT)$ ✓												
$X_{t,47}(ART)$												
$X_{t,48}(ROE)$							✓	✓				

Table 9 The relative optimal thresholds for Youden index c

Penalty function	Optimal threshold c			
	Shaoneng	Hengrui	Ploy	Foshan
LASSO	0.522	0.530	0.531	0.517
SCAD	0.483	0.525	0.413	0.512
MCP	0.487	0.544	0.507	0.506

Table 10 Hyper-parameter search space of machine learning

Methods	Hyper-parameter	Distribution
RF	N-estimators	List{300, 500, 800}
	Max-depth	Int[3, 30]
	Min-samples-split	Int[2, 20]
	Max-features	Uniform(0.3, 0.7)
SVM	Penalty-c	Uniform(1e-2, 1e4)
	Gamma	LogUniform(1e-4, 1e1)
	Kernel	fixed rbf
ANN	Hidden-layer-sizes	List{(64,),(128,),(256,),(128,64),(256,128)}
	L2-penalty-alpha	LogUniform(1e-6, 1e-2)
	Learning-rate-init	LogUniform(1e-4, 1e-1)

Table 11 Hyper-parameter search space of deep learning

Hyper-parameter	Fixed value	Search range
Learning-rate	1e-3	LogUniform(1e-3.3, 1e-2.7)
Dense-units	32	List{32, 64}
Dropout-rate	0.20	List{0.15, 0.20, 0.25}
Batch-size	16	List{16, 32}
Conv-filters	64	List{64, 80, 96}
LSTM-units	64	List{64, 80, 96}

deep learning methods with three types of indicators. Therefore, in this paper we propose a better prediction method to improve the accuracy of predicting weekly upward and downward stock price trends.

Compared with the existing LASSO/SCAD/MCP literature, the propose methods can not only validate prior findings, but also further identify sentiment indicators and fundamental indicators. Hu and Yang (2024a) primarily combined G-LASSO/G-SCAD/G-MCP with technical indicators, without considering the impact of fundamental indicators and sentiment indicators on stock price trends. The proposed methods demonstrate that, beyond technical indicators, both sentiment and fundamental indicators affect stock trends. This indicates that incorporating sentiment and fundamental indicators alongside technical indicators captures a more comprehensive set of factors affecting weekly stock price movements.

6.2 Discussion

In this paper we investigate LASSO/SCAD/MCP penalized logistic regressions with technical indicators, fundamental indicators, and sentiment indicators to pre-

Table 12 Random-search optimums for four datasets

Methods	Hyper-parameter	Optimums for Shaoneng	Optimums for Hengrui	Optimums for Poly	Optimums for Foshan
RF	N-estimators	800	800	800	800
	Max-depth	3	3	3	3
	Min-samples-split	15	15	15	15
	Max-features	0.8052	0.8052	0.8052	0.8052
SVM	Penalty-c	254.2013	254.2013	254.2013	254.2013
	Gamma	0.0003	0.0003	0.0003	0.0003
	Kernel	rbf	rbf	rbf	rbf
ANN	Hidden-layer-sizes	(128,)	(128,)	(64,)	(128, 64)
	L2-lenalty-alpha	0.0033	1.8179	2.7555	1.0072
	Learning-rate-init	0.0266	0.0772	0.0088	0.0004
DNN	Learning-rate	0.0005	0.0005	0.0007	0.0005
	Dense-units	64	64	32	64
	Dropout-rate	0.15	0.15	0.25	0.15
	Batch-size	16	16	16	16
CNN	Learning-rate	0.0007	0.001	0.001	0.0007
	Dense-units	32	32	32	32
	Dropout-rate	0.25	0.2	0.2	0.25
	Batch-size	16	16	16	16
	Conv-filters	96	64	64	96
LSTM	Learning-rate	0.001	0.001	0.001	0.001
	Dense-units	32	32	32	32
	Dropout-rate	0.2	0.2	0.2	0.2
	Batch-size	16	16	16	16
	LSTM-units	64	64	64	64

dict weekly stock price trends. In future research we can further explore other areas within the same framework, such as (i) commodity futures (ii) Bitcoin and other digital assets. Moreover, the sample is limited to four Chinese companies listed on the A-share market from 2010 to 2023. Future research focuses on the robustness property of the proposed methods in markets with greater institutional divergence, such as the U.S. and European stock markets. Finally, we can extend penalized logistic regression to penalized matrix/tensor logistic regression by introducing a predictive matrix/tensor based some different influencing factors. Furthermore, we employed the Youden index maximization method to determine the optimal classification threshold. In fact, the Youden index is not the only available threshold selection criterion, some other methods such as the Distance to Corner approach can also yield similar optimal thresholds (Staňková, 2023, 2025). In this paper we did not compare these alternative criteria. However, in future research we will further explore the practical impact of different threshold selection methods on model performance and stock price prediction.

Table 13 Predictive performance comparison

Data set	Penalty function	Sensitivity	Specificity	Precision	Accuracy
Shaoneng	LASSO	0.8137	0.6067	0.7034	0.7173
	SCAD	0.7255	0.7416	0.7629	0.7330
	MCP	0.7549	0.7079	0.7476	0.7330
	LASSO_24	0.7255	0.6966	0.7327	0.7120
	SCAD_24	0.7451	0.7079	0.7451	0.7277
	MCP_24	0.7647	0.6742	0.7290	0.7225
	RF	0.6180	0.7549	0.6875	0.6911
	SVM	0.6966	0.5490	0.5741	0.6178
	ANN	0.5843	0.6961	0.6265	0.6440
	DNN	0.6067	0.6961	0.6353	0.6545
	CNN	0.5882	0.5102	0.5102	0.5464
	LSTM	0.6941	0.5000	0.5463	0.5902
Hengrui	LASSO	0.7172	0.7604	0.7553	0.7385
	SCAD	0.7172	0.7708	0.7634	0.7436
	MCP	0.7475	0.7292	0.7400	0.7385
	LASSO_24	0.7475	0.7188	0.7327	0.7333
	SCAD_24	0.6869	0.7917	0.7727	0.7385
	MCP_24	0.6869	0.7917	0.7727	0.7385
	RF	0.6771	0.7475	0.7222	0.7128
	SVM	0.6667	0.6667	0.6598	0.6667
	ANN	0.5312	0.6667	0.6071	0.6000
	DNN	0.6979	0.5455	0.5982	0.6205
	CNN	0.4022	0.6632	0.5362	0.5348
	LSTM	0.5326	0.6316	0.5833	0.5829
Poly	LASSO	0.7890	0.6512	0.7414	0.7282
	SCAD	0.6239	0.8372	0.8293	0.7179
	MCP	0.7523	0.7093	0.7664	0.7333
	LASSO_24	0.5963	0.8721	0.8553	0.7179
	SCAD_24	0.6055	0.8605	0.8462	0.7179
	MCP_24	0.6055	0.8605	0.8462	0.7179
	RF	0.5814	0.7523	0.6494	0.6769
	SVM	0.6047	0.7431	0.6500	0.6821
	ANN	0.6860	0.5321	0.5364	0.6000
	DNN	0.6512	0.6881	0.6222	0.6718
	CNN	0.5854	0.6381	0.5581	0.6150
	LSTM	0.7683	0.4667	0.5294	0.5989
Foshan	LASSO	0.8515	0.5806	0.6880	0.7216
	SCAD	0.8515	0.5914	0.6935	0.7268
	MCP	0.8317	0.6022	0.6942	0.7216
	LASSO_24	0.8218	0.6022	0.6917	0.7165
	SCAD_24	0.8416	0.5914	0.6911	0.7216
	MCP_24	0.8416	0.5806	0.6855	0.7165
	RF	0.6667	0.7228	0.6889	0.6959
	SVM	0.6022	0.6832	0.6364	0.6443
	ANN	0.7204	0.6931	0.6837	0.7062
	DNN	0.5591	0.7822	0.7027	0.6753
	CNN	0.6703	0.4737	0.5495	0.5699
	LSTM	0.7363	0.3158	0.5076	0.5215

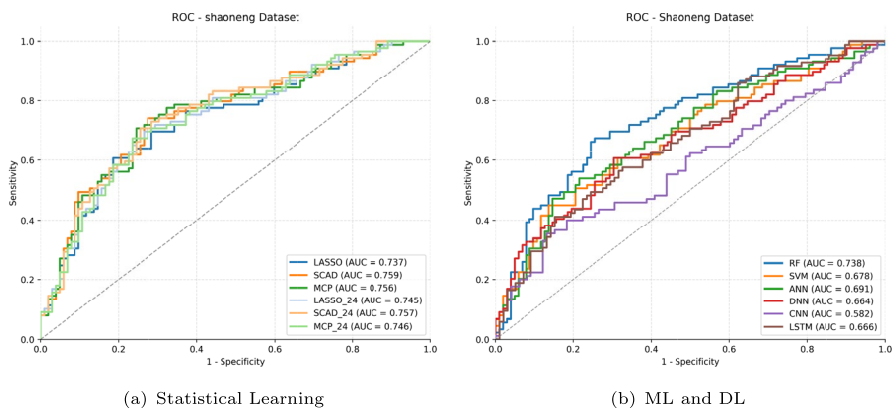


Fig. 11 ROC curves corresponding to the 12 methods for the Shaoneng

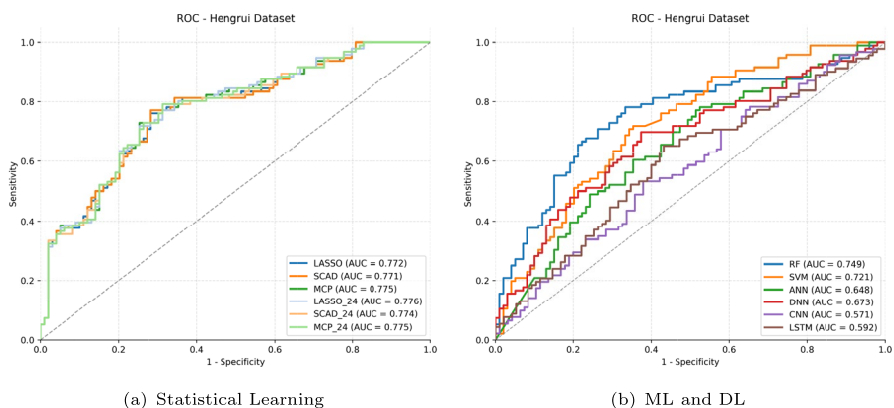


Fig. 12 ROC curves corresponding to the 12 methods for the Hengrui

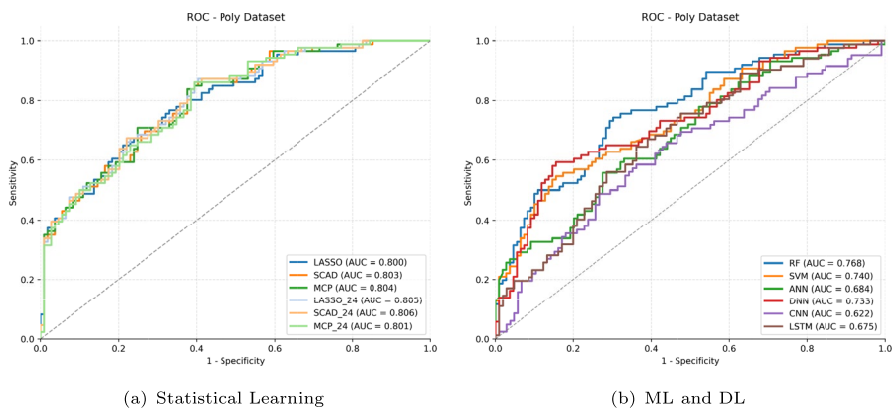


Fig. 13 ROC curves corresponding to the 12 methods for the Ploy

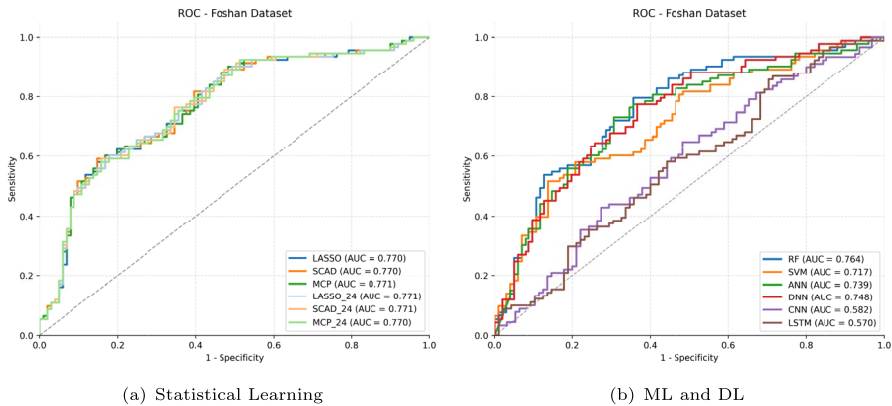


Fig. 14 ROC curves corresponding to the 12 methods for the Foshan

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Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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